Machine tests, maximum torque, generator mode

Cheat sheet · Day 11 · Tue 3 Nov · pack M8

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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INDUCTION MACHINE TESTS (per phase! V1 = VL/sqrt3, P1 = P/3)
 DC:        R1 per phase (line-line reading in star = 2 R1)
 Blocked:   s=1, ignore Rc, Xm.  R = P1/I1^2 = R1+R2'   ->  R2' = R - R1
            Q1 = sqrt((V1 I1)^2 - P1^2),  X = Q1/I1^2 = X1+X2',  X1 = X2' = X/2
 No-load:   s~0, ignore R1, X1.  Rc = V1^2/P1,  Xm = V1^2/Q1
            (Rc also absorbs friction + windage)
SIMPLIFIED CIRCUIT: Rc || jXm at terminals; series R1 + jX + R2'/s
 τ = 3 V1^2 (R2'/s) / ( ws [ (R1+R2'/s)^2 + X^2 ] )     ws = SYNCHRONOUS rad/s
MAX TORQUE:  s_τmax = R2' / |R1 + jX|,   n_τmax = (1 - s_τmax) nS
 τ_max = put s_τmax in τ formula  ( = 3V1^2 / (2 ws (R1 + |R1+jX|)) check )
 R2' changes s_τmax only, NOT τ_max
START (s=1): τ_st = 3V1^2 R2' / (ws[(R1+R2')^2+X^2]),  I_st ~ V1/|R1+R2'+jX|
R_EXT for τ_max at standstill:  R2' + R2EXT' = |R1 + jX|
V/f: V/f = VR/fR; scale V1, X, ws with f; R1, R2' fixed. τ_max ~const (R1->0)
GENERATOR: n > nS -> s<0, R2'/s<0, P_AG<0, τ<0, P_IN<0, Q_IN still >0
 shaft in = |P_AG|(1-s); elec out = |P_AG| - 3I^2 R1 - P_Fe; angle of I near 180 deg
SHORT ANSWERS: I_NL = magnetising (airgap) + core loss + friction/windage, pf low
 R2' = 0: no rotor loss, s_τmax = 0 (vertical like synch), zero starting torque
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